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Artificial creativity: can there be creativity without cognition?

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Abstract
This paper addresses the question of whether artificial systems can be considered creative in the absence of cognition. It advances a novel perspective by grounding the analysis on a foundational premise: generative AI systems, such as LLMs and GMIs, are non-cognitive. This distinction is established through the application of the Minimal Cognitive Grid (MCG), offering a more precise entry point into the creativity debate. Despite their non-cognitive nature, these systems produce outputs that meet standard criteria for creativity—novelty and usefulness—and reproduce, in functional terms, the stages of human creative processes. A comparative analysis with the Wallas–Jaoui model supports this claim. However, the absence of intentionality and authenticity limits any attribution of genuine creativity. So, how can we define Artificial Creativity? To resolve this, the paper introduces a minimal definition of artificial creativity as a non-cognitive, non-intentional, and non-authentic generative mechanism. This is the first attempt to define the concept directly, rather than by exclusion. The definition clarifies the theoretical boundaries between natural and artificial creativity, avoids anthropocentric bias, and establishes a foundation for future research in computational creativity and philosophy of AI.

Keywords Large Language Models · Creativity · Artificial Intelligence · Philosophy of Artificial Intelligence · Artificial Creativity

1 Introduction

In recent years, the question of whether Artificial Intelligence (AI) can be considered creative has gained increasing attention (Bown 2021; Bender et al., 2021; Kind 2022; Moruzzi 2021, 2025). This issue is central to our understanding of both AI and our relationship with intelligent machines, given that creativity has long been regarded as a defining trait of human thought (Arendt 2019; Boden 2004; Kant 2009; Heidegger 1991; Nietzsche 2017; Neumann 1999). Today, we are confronted with the unprecedented possibility that an artefact of human design might qualify as creative. The emergence of Generative Models for Images (GMIs) has transformed the field of Computational Creativity (Carnovalini and Rodà, 2020; Colton and Wiggins 2012; Lamb et al. 2018; Toivonen and Gross 2015). Some authors have even claimed that this development has redefined the landscape of creativity itself (Newton and Dhole 2023). AI systems—such as GMIs, Large Language Models (LLMs), and other generative tools (GenAI)—are now integrated into many stages of the creative process (Brock et al. 2018; Karras et al. 2019; Li and Sung 2021; Oh et al. 2018; Ramesh et al. 2022; Yee-King et al. 2024; Zhu et al. 2022). Earlier debates had already raised concerns about attributing creativity to machines, with some authors relegating it to the realm of "pseudo-creativity" (Cropley 1999; May 1959; Nicholls 1972). More recently, the term "artificial creativity" has gained traction to describe AI-generated outputs (Runco 2023a, b). Yet, the very distinction between "our" creativity and other forms raises an important question: are our definitions of creativity inherently anthropocentric? (Guckelsberger et al. 2017; Moruzzi 2021; Newell 1982; Newell et al. 1962). This issue becomes particularly salient in the context of animal cognition. Certain species, such as crows, demonstrate problem-solving abilities that might reasonably be described as creative (Moruzzi 2021; von Bayern et al. 2009, 2018). However, the forms of creativity expressed by non-human agents differ in essential ways from human creativity. Crows, for example, do not engage in something like artistic production. In contrast, artificial systems can now generate images that resemble art (Pedrazzi 2021), although they lack

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consciousness, emotional depth, and lived experience—factors often cited as prerequisites for genuine creativity (Hertzmann 2018; Runco 2023a, b). Other objections focus on the lack of autonomy and intentionality in AI systems (Moruzzi 2021), or argue that their outputs are merely the byproducts of formal algorithms (Kelly 2019). These are legitimate concerns. Nonetheless, some researchers maintain that AI systems meet the criteria of the so-called Standard Definition of Creativity (Runco and Jaeger 2012). Furthermore, empirical assessments using tools such as the Alternative Uses Test (AUT) and adapted versions of the Torrance Test of Creative Thinking (TTCT)—benchmarks for evaluating divergent cognition—suggest that AI systems can exhibit measurable creative performance (Guilford 1959, 1967, 1970; Torrance 1966; Goes et al. 2023). How, then, can we meaningfully speak of AI creativity? What kind of creativity is at stake, and how does it relate to human creativity? Is it productive to draw comparisons between the two, or should artificial creativity be treated as a distinct category? What are the theoretical and practical implications of such a distinction for the field of Computational Creativity? This study aims to clarify these questions through a structured analysis. Unlike previous approaches, it introduces a novel perspective: the investigation begins by establishing that current generative AI systems are non-cognitive. In Sect. 1, some preliminary distinctions are outlined. In Sect. 2, the Minimal Cognitive Grid (MCG) is applied to demonstrate that LLMs and GMIs lack the embodied and biological grounding typical of human cognition (Lieto 2021). This foundational claim, which has not previously been used as the starting point of an inquiry into AI creativity, provides a more robust theoretical basis for distinguishing artificial from human creativity. Section 3 offers a critical overview of the existing debate on AI and creativity, before turning in Sect. 4 to the question of whether certain AI systems instantiate mechanisms that resemble human creative processes. The Wallace–Jaoui model is employed as a neutral, pre-computational benchmark to facilitate this comparison. The analysis reveals that, although AI systems can replicate the stages of the creative process, they do so in a mechanistic, non-cognitive fashion. This reinforces the central argument while providing empirical grounding for the proposal. Section 5 introduces and defends a minimal and operational definition of "artificial creativity"—the first such definition in the literature. Rather than defining artificial creativity by contrast with human creativity, it is described as a non-cognitive, non-intentional, and non-authentic generative mechanism. This conceptualization offers a clear theoretical boundary and a structured foundation for future research. It provides a precise entry point into discussions on creative agency, authorship, and value in the age of generative systems. Moreover, although prior work has acknowledged that LLMs lack intentionality, this study offers a structured argument, based on the MCG, that definitively excludes the possibility of intentional states in such systems. The final section explores the broader implications of adopting a non-cognitive framework for artificial creativity, proposing future research directions and advocating for a more systematic and philosophically rigorous approach to the study of machine-generated creative output.

2 Some preliminary distinctions

As previously noted, a crucial but often overlooked point must be clarified. Most state-of-the-art generative models—textual, visual, or musical—are based on Transformer architectures (Vaswani et al. 2017), which in turn rely on Artificial Neural Networks (ANNs) (Russell and Norvig 2016). They consist of networks of simple data-processing units (neurons), connected by weighted links that modulate signal transmission based on statistical values (Cascini et al. 2010). Despite their biologically inspired nomenclature, ANNs lack fundamental features of human cognition and cannot be considered "cognitive" in the sense adopted by cognitive science (Lieto 2021). That's because during the development of AI as a discipline, two research programs emerged: one was "Machine-Oriented" AI, aimed at creating intelligent systems capable of solving complex problems; the other one was the "Human-Inspired" AI, which instead sought to develop programs inspired by human cognition. Although Machine-Oriented models, like LLMs, are now connectionist models, they cannot be cognitive because they have no structural analogies, more than a far inspiration, with our brain (Giunti et al., 2021; Shanahan 2023; Russell and Norvig 2016). LLMs are AI models designed for natural language understanding and generation. Examples such as ChatGPT or Google Gemini are based on connectionist models, specifically ANNs (Crick 1989; Brown et al. 2020; Radford et al. 2019). The very existence of algorithms such as the backpropagation, completely unrelated to the biological functioning of our neural system, should raise doubts about whether these systems can truly be compared to human cognition (Crick 1989; Hinton 2007; Lillicrap et al. 2020; Whittington and Bogacz 2019; Xiao et al. 2018).[1]

Similarly, the impressive results achieved by LLMs, the vast training data they require, the immense computational

[1] Some large language models incorporate training algorithms that have been described as, at least in part, biologically plausible. Nevertheless, the presence of even a single core mechanism that departs from biological plausibility substantially constrains the extent to which such systems can be legitimately characterised as cognitive. A more detailed discussion of this issue, and of its implications for the evaluation of cognitive architectures, is provided in the cited literature.

power involved, and the significant resource costs they incur serve as strong indicators against drawing direct comparisons between these systems and human cognition, in the sense of the MCG. (Caselli et al. 2023; Lieto et al 2018; Lieto 2021).

Why is it important to acknowledge that these potentially creative systems are not cognitive? Because if they were, we could not exclude the possibility that they possess forms of intentionality that we have yet to fully grasp. However, as will be discussed later, intentionality and authenticity are central components of human creativity—both of which seem to be absent in systems like LLMs. If these systems are not cognitive, they cannot be intentional or authentic: they remain functionalist rather than structuralist in nature. If their behavior were driven by intentions rather than by design, we would be compelled to attribute to them not only creativity, but also beliefs, desires, fears, and perhaps even consciousness. It is therefore essential to determine whether this is the case, before drawing further conclusions.[2] Some have argued that AI is not genuinely intelligent; rather, its most remarkable achievement lies in decoupling tasks from the intelligence traditionally required to perform them (Floridi 2019, 2023).

3 The minimal cognitive grid and the cognitive limits of LLMs

A useful framework for evaluating whether an AI system can be considered cognitive is the Minimal Cognitive Grid (MCG), developed by Antonio Lieto (2021, 2022). The MCG provides one of the few structured and theoretically grounded sets of criteria for assessing the cognitive plausibility of artificial systems. This paper represents the first systematic application of the MCG in a theoretical context explicitly aimed at excluding the cognitive status of LLMs. Before engaging with the MCG in detail, it is necessary to clarify a foundational distinction in AI theory: the one between structuralist and functionalist approaches. Functionalism, introduced in the philosophy of mind by Hilary Putnam in his 1960 essay “Minds and Machines”, maintains that mental states are defined by their causal-functional roles, independently of their physical or biological substrates (Putnam 1960). This distinction has direct implications for assessing the architecture and cognitive legitimacy of artificial systems—and thus for evaluating their potential creative status. In AI, this translates into the principle of *functional equivalence*: any system capable of replicating cognitive functions, regardless of its physical structure, could in principle be considered intelligent. Models such as IBM Watson and AlphaGo exemplify this approach (Russell and Norvig 2016).

However, under the criteria proposed by the MCG, these systems do not qualify as cognitive, as they lack explanatory power with respect to human cognition (Lieto 2021). Their outputs may resemble those of natural agents, but their internal processes diverge entirely from those found in human minds. By contrast, the structuralist approach emphasizes the alignment between the internal architecture of artificial systems and the structures observed in natural cognitive agents (Cordeschi 2002). This alignment not only facilitates technological development but also enables computational modeling as a method of scientific investigation in cognitive science. Yet, as Rosenblueth and Wiener famously observed, "The best material model of a cat is another, or preferably the same cat" (Rosenblueth, et al., 1945). That is, perfect replication is achievable only through identity, not analogy: an insight that underscores the limits of both approaches when pursued in isolation. A balanced perspective is therefore required. With this distinction in place, we can now turn to the Minimal Cognitive Grid to better understand the dimensions along which artificial systems may, or may not, approximate human cognition. The MCG provides a comparative framework for evaluating the structural and functional adequacy of AI models, thereby clarifying which systems can legitimately be described as cognitive. These dimensions include:

1. Balance between functionalism and structuralism. This criterion examines the relationship between functional and structural components in the design and implementation of artificial systems. The lower the ratio, the better.
2. Generality. How many activities can this model perform? In other words, how general is it? Excessive generality corresponds to a lower resemblance to human cognitive abilities.
3. Performance Match. How well does the model perform? According to Lieto, performance should be evaluated based on two factors: error analysis and execution time for a given task. Both of these aspects should ideally converge toward the characteristics typically observed in human cognition.

Let us apply the Minimal Cognitive Grid to a Large Language Model, taking GPT-4 as a representative case. From

[2] Some have questioned the necessity of emphasising the non-cognitive character of AI in discussions of creativity, arguing that cognition may be only one among several human capacities through which creativity is expressed, rather than a strict precondition for its emergence. Although this view deserves acknowledgement, the present analysis maintains that clarifying the cognitive status of AI systems remains essential. Without such clarification, there is a real risk of attributing to these systems qualities, such as intentionality and authenticity, that they do not in fact possess. The distinction is therefore substantive, not merely a matter of terminology.

the standpoint of the structuralist–functionalist distinction, GPT-4 is a purely functional system: it does not incorporate computational or memory constraints designed to emulate natural cognition. In this sense, it can be described as almost entirely functionalist. GPT-4 demonstrates a broad range of capabilities, as evidenced by numerous benchmarks (Bubeck et al. 2023; OpenAI 2023a, b; Petruzzellis et al. 2024). It performs effectively across various domains, including arithmetic, programming, and natural language processing—handling complex text analysis, summarization, and even multimodal tasks such as image generation and modification. Benchmarks consistently show its ability to complete complex tasks in minimal time.

A study conducted by Harvard Business School further supports this view, framing LLMs as knowledge and skill equalizers. The research highlights their impact on specific job sectors, showing how minimal human expertise can be compensated for by the use of such systems (Dell'Acqua et al. 2023). These findings indicate that LLMs can bypass the need for prior domain-specific training by directly enabling task completion. Some results also suggest that models like GPT-4 approach, and in some cases nearly surpass, human performance in Turing Test-like scenarios (Jones & Bergen 2024). While a full assessment would require a more detailed and systematic analysis, these considerations suffice to reasonably classify LLMs, including GPT-4, as non-cognitive systems. The same applies to other Transformer-based models. Most state-of-the-art Generative Models for Images (GMIs) are built upon Transformer architectures, typically integrating diffusion mechanisms. In such models, an input prompt is used to generate an image based on training data. The image is then incrementally degraded through the addition of noise until it becomes entirely unstructured; the model is trained to reverse this process and reconstruct a coherent output from the noisy data. Prominent examples include OpenAI's DALL·E, Google's Imagen, Midjourney, and Adobe Firefly (Borji 2022; Saharia et al. 2022; The Economist 2022).

In sum, most contemporary text-to-image systems, especially those used in artistic contexts, are based on LLMs or analogous Transformer-based architectures.

4 AI and creativity

Having provided clear evidence, for the first time in the literature on artificial creativity, that LLMs cannot be considered cognitive systems, we may now ask whether, in generating content, they engage a mechanism functionally comparable to that underlying human creativity. If such a mechanism exists, and if their outputs can reasonably be described as creative, then these systems instantiate a distinct, non-cognitive form of creativity that requires precise conceptual articulation. To assess the plausibility of this claim, a prior question must be addressed: how can creativity itself be recognized? Are there pre-generative AI frameworks capable of identifying a generalizable creative mechanism, one that could be meaningfully applied to artificial systems without assuming human cognition?

Various definitions and evaluative approaches have been proposed, focusing on aspects such as personality traits, achievements, actions, mindsets, or observable creative behaviors (Sternberg and Lubart 1999). Other models categorize creativity according to broader conceptual groupings—such as Person, Product, Process, and Press (Rhodes 1961), or the so-called 7Cs: Creators, Creating, Collaborations, Contexts, Creations, Consumption, and Curricula (Lubart 2017). A more outcome-focused definition, put forward by Boden (2004), frames creativity as the ability to produce ideas or artifacts that are new, surprising, and valuable. Some studies suggest that GenAI systems meet these criteria, producing outputs that are arguably novel, unexpected, and useful (Moruzzi 2021; 2025; Runco 2023a, b). A comprehensive review of the concept of creativity has already been conducted elsewhere (Moruzzi 2021, pp. 2–4)[3]; for this reason, we shall now turn to the next stage of the analysis.

For the purposes of this work, the 4P framework developed by Mel Rhodes appears most appropriate. While Boden's definition has been extensively discussed in relation to AI, and the models proposed by Sternberg and Lubart are less suitable for non-human agents, Rhodes' typology provides a structured and widely adopted framework that is more relevant for this context (Golann 1963; Jordanous 2016; Lamb et al. 2018; Cropley et al. 2023). The 4P model distinguishes among four dimensions of creativity:

- Person: The agent—human or non-human—considered as potentially creative. Theories in this area focus on traits or attributes associated with creative capacity.
- Process: The sequence of internal and external operations leading to the production of a creative output. Theories of process aim to describe the mechanisms underpinning creative activity.

[3] Additionally, the opening paragraphs of Still et al. (2019, pp. 219–222) provide a valuable introduction to the concept of creativity, particularly in relation to how it is taught and its implications for pedagogical theory. Their contribution is notable for framing creativity not only as a cognitive phenomenon but also as a practice shaped by educational structures and expectations—an angle that broadens the relevance of creativity studies beyond individual output. This pedagogical perspective complements the present investigation by highlighting how definitions of creativity are socially and institutionally embedded, and how these definitions influence both how creativity is cultivated and how it is assessed.

- Product: The artifact resulting from the creative process, judged for its originality and value. Theories of product focus on the criteria that render an outcome creative.
- Press: The cultural and environmental context that shapes, enables, and evaluates creativity. Theories of press explore how cultural factors influence the perception and validation of creativity (Lamb et al. 2018).

Among the four dimensions proposed by Rhodes, only Person, Process, and Product are consistently referenced in most theoretical models. In this context, the Person dimension is of limited relevance, as the creative status of AI systems is precisely the object of inquiry. Similarly, although the Press—the sociocultural environment—plays a significant role in shaping creativity, it is not central to this analysis, given the current openness and ambiguity of the debate. The Process dimension is also set aside here, as we have already established that AI systems lack cognitive processing; their creative functioning is therefore fundamentally distinct from that of humans. Nonetheless, in the following sections, we will introduce what we refer to as a mechanism—a concept distinct from process as defined by Rhodes (Cropley et al. 2023). This mechanism will serve as a useful analytical tool to identify a potential point of convergence between human and artificial creativity.

Accordingly, this study will focus on the Product dimension, assessing whether AI-generated outputs can be meaningfully regarded as creative. How, then, can the creativity of an AI-generated product be evaluated? Several established methods have been developed for assessing creative outputs. Among the most widely used are the Torrance Tests of Creative Thinking—that is used to evaluate divergent cognition—(TTCT; Lissitz and Willhoft 1985), the Runco Creativity Assessment Battery (rCAB; Runco 2011), the Consensual Assessment Technique (CAT; Amabile 1982), the Remote Associates Test (RAT; Mednick and Mednick 1967), and the Alternative Uses Test (AUT; Guilford 1967). Dietrich and Kanso (2010) argue that these assessment tools serve distinct purposes: the TTCT and AUT are primarily suited for evaluating divergent thinking, the CAT for assessing artistic and applied creativity, and the RAT for insight-based and convergent thinking. There is a lot of debate on these proposals. For example, the Amabile's CAT it's one of the golden standards of assessing creativity (Baer and McKool 2009) and is seen as the most popular method of assessing creative outputs (Kaufman et al. 2008). But for its focus on social context it's not useful in this discussion, like the "Press" dimension (Jeffries 2012). It is grounded in the straightforward premise that the most reliable way to evaluate the creativity of an artwork, theory, or artifact is through the judgment of domain-specific experts. In practice, creativity is rarely assessed through standardized checklists or general rubrics, whether in the context of prestigious awards such as the Nobel Prize or informal classroom competitions.[4]

A key distinction in creativity studies is between psychological creativity (P-creativity) and historical creativity (H-creativity) (Boden 2004; Miller 2019). The Consensual Assessment Technique (CAT) is better suited to evaluating H-creativity or "Big-C" creativity, while tests such as the Torrance Tests of Creative Thinking (TTCT) and the Alternative Uses Test (AUT) are more appropriate for assessing everyday, domain-general creativity—often referred to as "little-c" (Sternberg 2018; Kaufman et al. 2010) For our purpose the focus should be on psychological and everyday creativity, as said before, and TTCT could be thought as the primary assessment tool. The TTCT measures core creative abilities—fluency, flexibility, and originality—through domain-independent tasks such as Unusual Uses, Product Improvement, and Consequence Guessing.

Stated this, we should ask: could LLMs pass the TTCT test? This question rests on a mistaken premise: one does not "pass" the TTCT; rather, the test yields a score, typically expressed as a percentile based on the performance of other human participants. A study from Guzik et al. (2023) shows that GPT-4's performance in generating novel and unexpected ideas (originality) was noteworthy. The model outperformed human participants across various TTCT tasks, including "Just Suppose," which is designed to elicit imaginative thinking. Notably, 95% of GPT-4's responses in this task were rated as original, compared to only 24% from the control group. These results suggest that GPT-4 not only matches but may exceed human-level originality under current assessment criteria. While overall flexibility scores remain high, the model underperformed in specific TTCT tasks such as "Guessing Causes," "Guessing Consequences," and "Product Improvement." This may reflect limitations in prompt design, training data, or structural constraints inherent to the model. These outcomes highlight areas for potential improvement, particularly in enhancing the model's capacity to generate more diverse categories of ideas (ibid., 2023). However, the findings reported in this study raise some concerns, for instance, GPT-4's consistent scoring in the 99th percentile across most metrics strongly suggests prior exposure to TTCT content during training.

These results further raise critical questions about the adequacy of current creativity assessments. The TTCT,

[4] Some authors have grounded the concept of creativity in the foundational role of imagination (Kind 2022), or in the convergence of multiple evaluative and generative capacities (Taubitz et al. 2022). In discussions of artificial creativity, the emphasis on showing that generative systems are not cognitive aims precisely to exclude this dimension. Lacking a cognitive architecture, such systems cannot plausibly be regarded as possessing imagination in an intentional sense.

which primarily targets divergent thinking, appears insufficient to distinguish between genuinely creative processes and their functional imitations. Setting aside the conclusions of Guzik et al., it is evident that LLMs can reproduce the outputs of human divergent cognition without possessing the underlying cognitive mechanisms. This reinforces the necessity, clearly addressed through the MCG, of excluding machine cognition from the equation. Moreover, biases among human raters, such as a tendency to undervalue highly original or unfamiliar ideas (Licuanan et al. 2007; Blair and Mumford 2007), further complicate the reliability of such evaluations. In light of these issues, how can creativity be more accurately assessed in artificial systems?

In evaluating the creativity of a product, one practical approach is to examine whether a clear distinction can be drawn between AI-generated and human-created artifacts. Some argue that such a distinction is evident, asserting that only human beings have historically produced objects recognizable as "art" in the intentional and crafted sense (Hertzmann 2018, p. 20). However, recent developments complicate this claim. Since 2018, the auction house Christie's has sold artworks generated by artificial intelligence, signaling a shift in how such products are received and valued. One notable example is Portrait of Edmond de Belamy, created by the Paris-based collective Obvious using a Generative Adversarial Network (GAN), an early form of generative AI preceding current Transformer-based systems. Trained on a dataset of 15,000 portraits dating from the 14th to the twentieth century, the work was sold for $432,500 (Christie's, 12/12/2018). While market value is not a direct measure of artistic merit, such a transaction indicates that the work was perceived as sufficiently original and meaningful to merit high economic and symbolic recognition.[5] Another significant case is Neural Zoo by Sofia Crespo, a series of AI-generated images that blend elements of fauna and flora into surreal compositions. Using a dataset of natural imagery and a convolutional neural network, Crespo produces what she calls "speculative nature", synthetic forms that evoke natural life through technologically mediated abstraction (ArtNews, 28/12/2021). A third example is Pseudomnesia: The Electrician by Boris Eldagsen, which won the Creative Open Category at the Sony World Photography Awards in April 2023 (Glynn 2023). Upon receiving the award, Eldagsen declined it, revealing that the image had been generated using AI. His intention was to provoke reflection on the increasing indistinguishability between photographic realism and AI-generated imagery. The fact that the judges, unaware of its origin, selected the piece as the winner, seems to underscores a crucial point: AI-generated works can attain a level of aesthetic and conceptual sophistication comparable to that of human-created art. Taken together, these examples illustrate the growing difficulty—both perceptual and evaluative—of distinguishing between human and machine-generated creative artifacts. They challenge the assumption that artistic value must originate from human intentionality, and suggest that, at least in some domains, AI outputs are already integrated into systems of human cultural recognition.[6] Moreover, from a scientific standpoint, there is growing evidence that some generative AI systems exhibit creative capabilities. Haase and Hanel (2023) compared human-generated ideas with those produced by six Generative AI chatbots, alpha.ai, Copy.ai, ChatGPT (versions 3 and 4), Studio.ai, and YouChat. The quality and quantity of ideas were evaluated independently by both human judges and a trained AI model. The study found no significant qualitative difference between human and AI-generated outputs, although differences emerged in the mechanisms through which those ideas were produced. These findings reinforce the point previously discussed: generative systems can replicate creative outputs with high fidelity. However, this raises a critical question—central to the present analysis: even if the outputs are comparable, does the underlying creative mechanism coincide?

5 The creative mechanism

We now turn to the concept of the creative mechanism. While numerous contributions have addressed this construct, two influential models, developed prior to the advent of AI-generated imagery, remain particularly relevant. Notably, many subsequent accounts follow similar structural stages (De Giorgi 2016; Howard et al. 2008; Huang and Lee 2015). What distinguishes the present analysis is that, for the first time, the creative mechanism of artificial systems is examined through theoretical models that predate the emergence

[5] The commercial success of AI-generated artworks does not, in itself, reveal the evaluative intentions of those who purchase them. Such transactions may reflect speculative or strategic behaviour, rather than genuine artistic appreciation. Despite this, Christie's auction, when considered alongside other indicators, offers a strong signal of emerging social recognition: that some AI-generated products are perceived, at least by a portion of the public, as possessing artistic value. While not conclusive, this constitutes a significant piece of empirical evidence within the wider argument.

[6] Some studies suggest that expert artists are still able to distinguish with a fair degree of accuracy between machine-generated and human-made artworks (Ha et al. 2024). The reliability of this distinction, however, remains uncertain. The study in question is constrained by the size and diversity of its dataset, making it difficult to generalise the findings. For present purposes, it is not necessary to show that AI-generated works are readily identifiable as such, but rather that they display a level of artistic value sufficient to warrant consideration within creative domains.

of generative AI. This methodological choice is not merely historical; it provides a neutral, cognitively grounded baseline against which to assess the structure of AI-generated creative processes. This constitutes one of the central original contributions of the paper. The two models considered are Graham Wallas's The Art of Thought (Wallas 1926) and Hubert Jaoui's La Créativité: Mode d'Emploi (Jaoui, 1994). Despite their temporal distance, both propose comparable sequences of stages in the creative process, which justifies addressing them in parallel. As will be shown, their core stages are echoed across several major theoretical frameworks in the creativity literature.

Let's now look at the combined characteristics of the Wallas-Jaoui creative mechanism:

- Impregnation (or preparation): The creator gathers as much data, observations, and information as possible, absorbing it like a sponge, becoming aware of the default situation, and setting a goal.
- Incubation: The time spent allowing knowledge to settle.
- Illumination: The moment when the idea emerges, and like an epiphany, it must be materialized.
- Verification: A revision of what has been done to ensure that it is "beautiful" or "good," at least according to the chosen standards.

Now, from an initial analysis, it will be clear how the AI generation process aligns with these aspects. The first phase, impregnation, as well as the second, incubation, is not dissimilar to what happens in Machine Learning: the model is trained on specific datasets, which can be broader or narrower depending on the desired outcome for the machine. For instance, consider the work of Sophia Crespo, where the CNN she selected is trained on a specific dataset to achieve the chosen creative output. The creator, or in this case, the generator, gathers observable data and the information associated with it, often compiled into textual labels linked to training images. The goal set, namely generation, is the intrinsic purpose of designing such a model. The *preparation* phase can be understood as the capacity for *problem recognition*—the ability to identify a problem when it arises. While some scholars argue that this is a uniquely human faculty (Cropley et al. 2023), others have highlighted the capacity of generative AI to detect inconsistencies in reasoning, evaluate problematic situations, and adapt accordingly. Notably, the ability to identify and address problems has been a focus of artificial systems since early models such as the General Problem Solver (GPS) developed by Newell and Simon, which aimed to formalize problem-solving as a core feature of machine intelligence (Newell and Simon 1961; 1972; Russell and Norvig 2016).[7]

In models based on Transformer architecture, the incubation phase appears to be subsumed within the preparation phase. Processing occurs primarily during the training stage and does not require a period of "settling," but rather of "testing." Through supervised learning, the model learns to interpret and apply its internal representations. This process is not unlike that of an aspiring painter who begins producing work based on prior study and the materials previously collected. This pattern holds across Transformer-based models.

The illumination phase corresponds to the moment of activation by a prompt or input code during the generation process. The model receives an instruction that catalyzes the activation of its learned parameters, producing an output consistent with its training data. One might argue that this moment is not truly spontaneous, since it depends on human initiation. However, this line of reasoning would require assuming that artistic ideas emerge spontaneously in the human mind, a claim difficult to defend. In many historical and contemporary cases, artistic "epiphanies" are the result of interaction: inspiration, commissions, external suggestions, and dialogue. In this sense, prompts may be seen as analogous to requests or stimuli that guide human creativity, a notion consistent with combinatorial models of creativity (Koestler 1964; Bruner 1979; Boden 2004; Dietrich 2015; Wiggins 2019). Even accepting the premise that prompting is essential, it would still be inaccurate to claim that generative models produce output only in response to external commands. During unsupervised training, AI systems generate outputs from internally generated prompts or labels designed to facilitate learning—for instance, prompts related to style, subject, or context. The case of Generative Adversarial Networks (GANs) is illustrative: one network (the generator) produces outputs, while a second network (the discriminator) evaluates them against the training data. When discrepancies are detected, a backpropagation algorithm updates the generator, and the process repeats. This iterative exchange simulates an evaluative loop.

More complex models, such as Mixtures of Experts (MoEs), involve multiple neural networks with specialized functions that collaborate to produce a single output. In a MoE configuration involving GANs or recurrent networks, one group may generate images while another produces internal prompts to guide the generation process. This results

[7] However, some scholars have argued that the heuristics proposed by Newell and Simon are insufficient to support the claim that the General Problem Solver genuinely employed creativity in producing novel solutions. For a more detailed critique, see Ohlsson (2012).

in a system capable of autonomously producing potentially creative artifacts (Cai et al. 2024).

The verification phase occurs both internally—through discriminator feedback during training—and externally, through human evaluation, as seen in models like GPT-4 or Adobe Firefly. In these cases, generated outputs are assessed by end-users or prompt-creators, introducing a form of social feedback. The discriminator-generator loop in GANs can be interpreted as a functional analogue of human trial-and-error processes (Elgammal et al. 2017; Moruzzi 2021). In light of these observations, it may be argued that the creative mechanism underlying human artistic production shares functional similarities with that employed by Transformer-based AI models. This should not be unexpected, given that such models were designed based on human strategies for solving complex problems. While not cognitive in nature, these machine-oriented systems reproduce patterns of human problem-solving—including those involved in creative tasks.[8] Suspending the ontological question of creativity, it is reasonable to conclude that the underlying mechanism of generation is, at least in functional terms, comparable.

6 AI is not truly creative

In summary, three relevant points have been put on the table: (1) LLMs and the most widely used generative AI systems cannot be considered cognitive; (2) their outputs are at human-level creativity or above, and are often attributed artistic value by human evaluators—sometimes even being mistaken for human-made works; (3) these systems appear to exhibit a creative mechanism that closely resembles, or at least aligns with, general models of the creative process originally developed to describe human creativity. Returning to the central question: can generative AI systems be conceived as truly creative? The answer appears to remain negative. While they may replicate or simulate certain functional aspects of creativity, the absence of cognition, intentionality, and subjective experience limits the extent to which their outputs can be considered genuinely creative in the full, human sense of the term (Moruzzi 2025; Runco 2023a, b). Until 2023, the standard theory of creativity—centered on the criteria of novelty and usefulness—did not offer a definition sufficiently robust to exclude AI-generated outputs from the domain of human creativity (Runco & Jaeger 2012). It is now widely acknowledged that these two characteristics can be attributed to the outputs of generative AI systems (Guzik et al. 2023; Moruzzi 2021; Runco 2023a, b). However, the updated formulation of the theory, explicitly revised to address this issue, introduces a distinction that excludes what the author terms "pseudo-creativity" or "artificial creativity" (Runco 2023a). As previously discussed, despite functional similarities between human and AI-generated creative processes, current AI systems lack two fundamental elements in the revised model: intentionality and authenticity. What is particularly noteworthy, however, is that it has become necessary to revise the standard definition of creativity in order to exclude artificial systems. This shift may be interpreted as indirect evidence of the creative potential of such systems: the earlier, widely accepted definition—used for over a decade—was insufficient to distinguish between human and non-human creativity. This observation underscores the need for a distinct definition of artificial creativity, one capable of acknowledging the specific nature of AI-generated outputs while avoiding the persistence of anthropocentric assumptions that continue to shape creativity studies (Guckelsberger et al. 2017; Moruzzi 2021; Newell 1982; Newell et al. 1962).

Despite the originality and usefulness of AI-generated outputs, they cannot be considered truly original, as they lack two essential components of creativity: choice and intention (Kind 2022; Runco 2023a, b). According to Runco, what appears as AI creativity is better understood as a mistaken attribution, analogous to the attribution of mental states or intentionality to systems such as GPT-4 (Goldstein and Levinstein 2024). Here, intentionality refers to the capacity of mental states to be directed toward or represent objects, properties, or states of affairs (Jacob 2023). Because the output resembles that of human creativity, the mechanism appears comparable, and the potential seems present, observers are prone to interpret these traits as evidence of creativity. However, in the absence of desires, beliefs, emotions, or other mental states, the creativity exhibited by artificial systems cannot be equated with that of human agents. To clarify the concept of authenticity, it is useful to refer to the Stanford Encyclopedia of Philosophy, which defines the authentic as "faithful to the original," and describes the authentic individual as one who acts in accordance with

[8] It may be objected that, within the described mechanism, human involvement is required at every stage, and that the only step performed "independently" by a large language model, once prompted, is the generation of alternatives. This objection is not decisive. While it is true that a human must initiate the system, such dependence does not negate the autonomy of subsequent stages. In unsupervised learning contexts, the impregnation and preparation phases can proceed without direct human intervention. Incubation is embedded within the training architecture itself, illumination may be triggered by internally generated or random prompts, and verification can be carried out autonomously, particularly in architectures such as GANs or MoEs. Taken together, this suggests that the entire mechanism can unfold without continuous human oversight. What remains distinctive is that artificial creativity lacks intentionality: only human creativity can arise without external prompting, precisely because it is rooted in intentional states. The point of the analysis is not to equate the ontological bases of human and artificial creativity, but to show that the observable mechanism of generative production follows a structurally similar sequence.

their own desires, motivations, ideals, or beliefs (Varga et al. 2023). Once again, the notion of intentionality and the presence of mental states becomes central, further exposing the limitations of artificial creativity. The inability of AI systems to possess mental states and the content they entail appears to be the fundamental reason for their lack of authentic creativity. As David Means observed in a 2023 article for *The New York Times*:

> I'm not going to go pick a fight with A.I., or even argue with the fact that this technology can mimic artwork, or assist humans in the creation of art, but I can say... that A.I. will never be able to do what I can do because A.I. has never felt what I've felt. It will never move through the emotional matrix of living a singular, individual life (Means 2023).

In summary, while the generative mechanism behind image production may appear structurally analogous in humans and AI systems, the underlying dynamics and foundational criteria for defining creativity clearly exclude Transformer-based architectures from being considered truly creative. This, however, is not unexpected. From the outset, it has been acknowledged that these generative models are machine-oriented and make no claim to replicate human cognition. Their divergence from human cognitive traits is therefore unsurprising. In this regard, Runco consistently refers to the output of such systems as examples of "pseudo-creativity" or "artificial creativity." (Runco 2023a, b) Nonetheless, it has been demonstrated that the stages of the creative mechanism—as modeled in pre-Generative AI theories—are mirrored in the functioning of these systems. Moreover, the outputs they produce often possess artistic value, sometimes indistinguishable from human-made works, as illustrated by the case of Pseudomnesia: The Electrician. Perhaps most strikingly, one of the most widely cited theories of creativity required reformulation precisely because it failed to distinguish between human and AI creativity on conceptual grounds, despite clear intuitive and experiential differences. In light of these considerations, we are now in a position to propose a first, well-grounded definition of artificial creativity:

Artificial creativity is the generative mechanism, not intentional and not authentic, expressed by non-cognitive artificial systems.

The proposed definition may be regarded as a minimal and sufficient account of artificial creativity, though not exhaustive. As such a definition has not been explicitly formulated in previous literature, it will require further theoretical refinement and empirical investigation. Nevertheless, it offers a promising and conceptually rigorous foundation for future research. The originality of the definition lies in its grounding: by employing the Minimal Cognitive Grid, the attribution of intentionality to these systems is definitively ruled out. This exclusion enables a novel and analytically precise formulation of artificial creativity.

The notion of "generative mechanism" referenced here corresponds to the structure previously discussed, and should be understood as a general, procedural account of how non-cognitive systems can produce creative outputs. Crucially, this mechanism is defined as non-intentional and non-authentic. Unless these two features are explicitly excluded, any philosophical attempt to distinguish artificial creativity from human creativity remains ambiguous and unstable. The proposed definition focuses exclusively on the generative mechanism itself, eschewing psychological, contextual, and intentional dimensions. Unlike human creativity, which may involve factors such as personal identity (e.g., genius), sociocultural influence (press), or deliberate expressive intent, artificial creativity is here defined in strictly mechanistic and structural terms. This conceptual restriction ensures definitional clarity and prevents conflation between simulated creativity and creativity proper.

Adopting this minimal definition, artificial creativity as a non-cognitive, non-intentional, non-authentic generative mechanism, offers several theoretical and methodological advantages. First, it enables a clear delineation of the phenomenon, avoiding anthropomorphic projections and conceptual ambiguity. Second, it facilitates the development of operational evaluation criteria based solely on observable outputs, without presupposing mental states in artificial systems. This supports integration with empirical methodologies in AI and cognitive science. Furthermore, the definition provides a coherent foundation for a structured framework in artificial creativity, where the distinction between human and artificial creativity is embedded at the architectural level. Emphasizing the cognitive status of the system—rather than the aesthetic or functional quality of its output—aligns with current research in computational cognitive science and avoids speculative claims about artificial minds. By abstaining from normative and anthropocentric assumptions, the definition establishes a stable conceptual platform for the responsible design and evaluation of generative systems. It contributes meaningfully to ongoing debates about authorship, agency, and creative value in both artistic and industrial domains. For instance, domains such as creative feedback stand to benefit from this clarification. If artificial creativity is defined without cognition, intentionality, or authenticity, then it becomes evident why AI cannot support a Deweyan model of creative feedback. As Still et al. (2019) argue, creative feedback is inherently relational and presupposes mutual understanding, capacities that current artificial systems lack. The field of Explainable Computational Creativity (XCC) could also benefit from adopting this definition. If creativity in AI is reduced to a generative process, then current LLMs are insufficient as foundations for XCC. Core design principles of XCC—such as mental models, memory,

argumentation, and process transparency—presuppose cognitive functions that extend beyond generation. This definition therefore clarifies that XCC requires distinct architectures. Accordingly, collaborations envisioned by XCC differ in kind from those enabled by standard generative models (Llano et al. 2022).[9] Future research may explore hybrid configurations in which human agents are embedded in the generative loop. As Moruzzi (2021) suggests, such configurations may warrant attributing limited autonomy or naïveté to the system as a whole, supporting a more nuanced taxonomy of creativity across human–machine assemblages. However, under the framework proposed here, full human autonomy is not a necessary condition for artificial creativity. As defined, artificial creativity refers exclusively to a non-cognitive, non-intentional generative mechanism, irrespective of whether human input is present.

7 Conclusion

The analysis proposed here has aimed to conceptually delineate artificial creativity as a generative mechanism independent of cognition, intentionality, and authenticity. Rather than offering a merely negative critique of current generative systems, this definition introduces a functional and theoretically grounded framework that distinguishes structurally distinct modes of creative production. In doing so, it contributes to a broader reconceptualization of creativity as a plural and stratified phenomenon—one that includes, but is not exhausted by, human cognition. Far from diminishing the value of artificial output, this framework enhances our ability to understand how creativity can manifest in systems that lack the evaluative, autobiographical, and social scaffolding characteristic of human imagination (Taubitz et al. 2022). As Llano et al. (2022) observe, co-creative and explainable computational systems require communicative and memory structures that current LLMs do not possess. This reinforces the claim that while LLMs may participate in creative workflows, they do not enact creativity in the strong cognitive sense.

Nevertheless, as Moruzzi (2021) suggests, hybrid human–machine configurations may complicate this taxonomy, allowing for forms of partial autonomy or naïveté to emerge at the system level. Within this context, the definition proposed here functions as a conceptual boundary that can be flexibly applied across domains—from artistic production to human–computer interaction—while preserving theoretical clarity. It also enables a more rigorous and less anthropocentric comparison between natural and artificial forms of creativity, as advocated by Jordanous (2012). Crucially, the originality of this work lies in grounding the entire discussion in the prior exclusion of cognition via the Minimal Cognitive Grid, and in offering the first direct, positive definition of artificial creativity—not merely by contrast with human creativity, but as a self-contained construct. This approach clarifies longstanding ambiguities in the literature and provides a conceptual starting point for future research. Rather than replicating human creativity, the definition makes its difference explicit. It enables a more precise mapping of what generative systems can and cannot do, while simultaneously enriching our understanding of human creativity by contrast. Future research may build on this conceptual foundation to refine evaluation criteria, develop hybrid systems, or inform the design of architectures that aim not merely to generate content but to engage in structured creative collaboration. In this sense, artificial creativity, as defined here, is both a conceptual limit and a productive frontier. We have developed systems that, while lacking cognitive faculties, exhibit a form of creativity substantial enough to provoke revisions of long-standing theoretical assumptions. This opens a series of foundational questions: Should creativity research adopt a less anthropocentric, more inclusive framework capable of accommodating artificial as well as natural forms of creativity? Might classical models of the creative process, developed prior to generative AI. be reconceptualized as abstract schemas, applicable beyond human cognition? Could artificial creativity, precisely due to its structural limitations, serve as a contrasting lens through which to better understand the cognitive, intentional, and social dimensions of human creativity?

Moreover, to what extent should the artistic or epistemic value of AI-generated works be reassessed, not merely in light of their increasing presence, but in view of their detachment from autonomous agency or evaluative understanding? How should we distinguish between systems that simulate creativity and those that participate in co-creative processes, especially when users increasingly attribute value to outputs regardless of their origin? Finally, what are the theoretical and practical consequences of recognizing artificial creativity as a distinct category? If current systems excel in fluency, flexibility, and even originality, but remain unable to evaluate the relevance or utility of their own productions, what does this asymmetry reveal about the relationship between generation and judgment? And what normative criteria should guide future classifications as generative systems continue to evolve? These questions remain open, but engaging with them from the conceptual foundation proposed here may sharpen our understanding of both artificial creativity and the broader architecture of creative agency.

[9] That is, it is not merely sufficient to render some LLMs explainable in order to qualify them for participation in the XCC paradigm. Explanation alone does not supply the cognitive grounding required by the XCC design principles.

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Declarations

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References

Amabile TM (1982) Social psychology of creativity: a consensual assessment technique. J Pers Soc Psychol 43(5):997–1013. https://doi.org/10.1037/0022-3514.43.5.997

Arendt A (2019) The human condition. The University of Chicago Press

Artnews.com (2021) Sofia Crespo's speculative nature. Retrieved from: https://www.artnews.com/art-in-america/features/sofia-crespo-speculative-nature-1234614511/

Baer J, McKool S (2009) Assessing creativity using the consensual assessment. In: Schreiner C (ed) Handbook of assessment technologies, methods, and applications in higher education. IGI Global, pp 65–77. https://doi.org/10.4018/978-1-60566-667-9.ch004

Bender EM, Gebru T, McMillan-Major A, Mitchell M (2021) On the dangers of stochastic parrots: can language models be too big? Proceedings of the 2021 ACM conference on fairness, accountability, and transparency, 610–623

Blair CS, Mumford MD (2007) Errors in idea evaluation: preference for the unoriginal? J Creat Behav 41(3):197–222. https://doi.org/10.1002/j.2162-6057.2007.tb01288.x

Boden MA (2004) The creative mind: myths and mechanisms. Routledge, London

Borji A (2022) Generated faces in the wild: quantitative comparison of stable diffusion, midjourney and DALL-E 2. https://arxiv.org/abs/2210.00586

Bown O (2021) Beyond the creative species: making machines that make art and music. MIT Press, Cambridge, MA

Brock A, Donahue J, Simonyan K (2018) Large scale GAN training for high fidelity natural image synthesis. https://arxiv.org/abs/1809.11096

Brown T et al (2020) Language models are few-shot learners. In: Advances in neural information processing systems 33 (NeurIPS 2020)

Bruner J (1979) On knowing: essays for the left hand. Belknap Press

Bubeck S, Chandrasekaran V, Eldan R, Gehrke J, Horvitz E, Kamar E, Lee P, Lee YT, Li Y, Lundberg S, Nori H, Palangi H, Ribeiro MT, Zhang Y (2023) Sparks of artificial general intelligence: early experiments with GPT-4. https://doi.org/10.48550/arXiv.2303.12712

Cai W, Jiang J, Wang F, Tang J, Kim S, Huang J (2024) A survey on mixture of experts. arXiv:2407.06204

Carnovalini F, Rodà A (2020) Computational creativity and music generation systems: an introduction to the state of the art. Front Artif Intell. https://doi.org/10.3389/frai.2020.00014

Caselli T, Lieto A, Nissim M, Patti V (2023) Sono solo parole. ChatGPT: anatomia e raccomandazioni per l'uso. Sistemi intelligenti 35(2):307–320. https://doi.org/10.1422/10813

Cascini GL, Donato V, Tamburrini O (2010) Introduzione alle reti neurali. Elementi di informatica in diagnostica per immagini. Springer, Milan, pp 217–233

Christies.com (2018, Dec. 12) Obvious and the interface between art and artificial intelligence. Retrieved from https://www.christies.com/en/stories/a-collaboration-between-two-artists-one-human-one-a-machine-0cd01f4e232f4279a525a446d60d4cd1

Colton S, Wiggins GA (2012) Computational creativity: the final frontier? In: Proceedings of the 20th European conference on artificial intelligence (ECAI 2012). IOS Press, pp 21–26. https://doi.org/10.3233/978-1-61499-098-7-21

Cordeschi R (2002) The discovery of the artificial: behavior, mind and machines before and beyond cybernetics. Kluwer Academic Publishers, Dordrecht

Crick F (1989) The recent excitement about neural networks. Nature 12(6203):129–132. https://doi.org/10.1038/337129a0

Cropley AJ (1999) Creativity and cognition: producing effective novelty. Roeper Rev 21(4):253–260. https://doi.org/10.1080/02783199909553972

Cropley DH, Medeiros KE, Damadzic A (2023) The intersection of human and artificial creativity. In: Henriksen D, Mishra P (eds) Creative provocations: speculations on the future of creativity, technology & learning. Springer, pp 19–34. https://doi.org/10.1007/978-3-031-14549-0_2

De Giorgi L (2016) La creatività: alcune prospettive per un costrutto difficile da definire. In: Mizar, Costellazione di pensieri, n. 2–3

Dell'Acqua F, McFowland III E, Mollick ER, Lifshitz-Assaf H, Kellogg K, Rajendran S, Krayer L, Candelon F, Lakhani KR (2023) Navigating the jagged technological frontier: field experimental evidence of the effects of AI on knowledge worker productivity and quality (Harvard Business School Technology & Operations Mgt. Unit Working Paper No. 24-013; The Wharton School Research Paper). https://doi.org/10.2139/ssrn.4573321

Dietrich A, Kanso R (2010) A review of EEG, ERP, and neuroimaging studies of creativity and insight. Psychol Bull 136(5):822–848. https://doi.org/10.1037/a0019749

Dietrich E (2015) Excellent beauty: the naturalness of religion and the unnaturalness of the world. Columbia University Press

Elgammal A, Liu B, Elhoseiny M, Mazzone M (2017) CAN: creative adversarial networks, generating "art" by learning about styles and deviating from style norms. In: Proceedings of the 8th international conference on computational creativity (ICCC 2017). Georgia Institute of Technology

Floridi L (2019) What the near future of artificial intelligence could be. Philos Technol 32(1):1–15. https://doi.org/10.1007/s13347-019-00345-y

Floridi L (2023) The ethics of artificial intelligence: principles, challenges, and opportunities. Oxford University Press

Gabora L, Kaufman SB (2010) Evolutionary perspectives on creativity. In: Kaufman J, Sternberg R (eds) The Cambridge handbook of creativity. Cambridge University Press, Cambridge, UK, pp 279–300

Giunti M, Garavaglia G, Pinna S (2021) Menti e Macchine. Teorie scientifiche e filosofiche. 2021, Le Monnier

Glynn P (2023) Sony world photography award 2023: winner refuses award after revealing AI creation. BBC News. https://www.bbc.com/news/entertainment-arts-65296763

Góes LF, Volpe M, Sawicki P, Grses M, Watson J (2023) Pushing GPT's creativity to its limits: alternative uses and torrance tests. In: Proceedings of the 14th international conference for computational creativity. University of Leicester, Ontario, Canada, pp 1–6

Golann SE (1963) Psychological study of creativity. Psychol Bull 60:548–565. https://doi.org/10.1037/h0041573

Goldstein S, Levinstein BA (2024) Does ChatGPT have a mind? https://arxiv.org/abs/2406.14892

Guckelsberger C, Salge C, Colton S (2017) Addressing the "Why?" in computational creativity: a non-anthropocentric, minimal model of intentional creative agency. In: Proceedings of the 8th international conference on computational creativity (ICCC 2017). Association for Computational Creativity. http://research.gold.ac.uk/id/eprint/21743

Guilford JP (1959) Traits of creativity. In: Anderson HH (ed) Creativity and its cultivation. Harper & Row, New York, pp 142–161

Guilford JP (1967) Creativity: yesterday, today, and tomorrow. J Creat Behav 1(1):3–14. https://doi.org/10.1002/j.2162-6057.1967.tb00002.x

Guilford JP (1970) Creativity: retrospect and prospect. J Creat Behav 4(3):149–168. https://doi.org/10.1002/j.2162-6057.1970.tb00856.x

Guzik EE, Byrge C, Gilde C (2023) The originality of machines: AI takes the Torrance test. J Creat 33(3):100065. https://doi.org/10.1016/j.yjoc.2023.100065

Ha AYJ, Passananti J, Bhaskar R, Shan S, Southen R, Zheng H, Zhao BY (2024) Organic or diffused: can we distinguish human art from AI-generated images? https://doi.org/10.48550/arXiv.2402.03214

Haase J, Hanel PHP (2023) Artificial muses: generative artificial intelligence chatbots have risen to human-level creativity. J Creat 33(3):100066. https://doi.org/10.1016/j.yjoc.2023.100066

Hinton GE (2007) How to do backpropagation in a brain [presentation]. In: Deep learning workshop. NIPS 2007. https://www.cs.toronto.edu/~hinton/backpropincortex2007.pdf

Howard TJ, Culley SJ, Dekoninck EA (2008) Describing the creative design process by the integration of engineering design and cognitive science methodologies. Des Stud 29(2):160–180. https://doi.org/10.1016/j.destud.2008.01.001

Heidegger M (1991) Nietzsche, vol 1. The Will to Power as Art, Harper One

Hertzmann A (2018) Can computers create art? Arts 7(2):18. https://doi.org/10.3390/arts7020018

Huang X-H, Lee JC-K (2015) Disclosing Hong Kong teacher beliefs regarding creative teaching: five different perspectives. Think Skills Creat 15:37–47. https://doi.org/10.1016/j.tsc.2014.11.003

Jacob P (2023) Intentionality. In: Zalta EN (ed) The Stanford Encyclopedia of philosophy (Spring 2023 Edition). Retrieved from https://plato.stanford.edu/entries/intentionality/

Jaoui H (1994) La créativité mode d'emploi. Connaissance du problème, applications pratiques. ESF éditeur, Montrouge

Jeffries KK (2012) Amabile's consensual assessment technique: Why has it not been used more in design creativity research? In: Proceedings of the 2nd international conference on design creativity (ICDC2012). The Design Society, Glasgow, UK, pp 211–220

Jones C, Bergen B (2024) Does GPT-4 pass the turing test? In: Duh K, Gomez H, Bethard S (eds) Proceedings of the 2024 conference of the North American chapter of the association for computational linguistics: human language technologies (volume 1: long papers). Association for Computational Linguistics, pp 5183–5210. https://aclanthology.org/2024.naacl-long.290/

Jordanous AK (2012) A standardised procedure for evaluating creative systems: Computational creativity evaluation based on what it is to be creative. Cogn Comput 4(3):246–279. ISSN 1866-9956

Jordanous A (2016) Four PPPperspectives on computational creativity in theory and in practice. Connect Sci 28:194–216. https://doi.org/10.1080/09540091.2016.1151860

Kant I (2009) Critique of judgement. Oxford University Press

Karras T, Laine S, Aila T (2019) A style-based generator architecture for generative adversarial networks. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR 2019). IEEE, pp 4401–4410. https://doi.org/10.1109/CVPR.2019.00453

Kaufman JC, Plucker JA, Baer J (2008) Essentials of creativity assessment. Wiley, Hoboken, NJ US

Kelly SD (2019) A philosopher argues that an AI can't be an artist. MIT Technol Rev. https://www.technologyreview.com/s/612913/a-philosopher-argues-that-an-ai-can-never-be-an-artist/

Kind A (2022) Imagination and creative thinking. Cambridge University Press.https://doi.org/10.1017/9781108973335

Koestler A (1964) The act of creation. Hutchinson & Co., London, UK

Lamb C, Brown DG, Clarke CLA (2018) Evaluating computational creativity: an interdisciplinary tutorial. ACM Comput Surv 51:1–34. https://doi.org/10.1145/3167476

Li S, Sung Y (2021) Inco-gan: variable-length music generation method based on inception model-based conditional GAN. Mathematics 9(4):387. https://doi.org/10.3390/math9040387

Lieto A, Lebiere C, Oltramari A (2018) The knowledge level in cognitive architectures: current limitations and possible developments. Cogn Syst Res 48:39–55. https://doi.org/10.1016/j.cogsys.2017.05.001

Lieto A (2021) Cognitive design for artificial minds. Routledge, New York

Lieto A (2022) Analyzing the explanatory power of bionic systems with the minimal cognitive grid. Front Robot AI 9:888199. https://doi.org/10.3389/frobt.2022.888199

Licuanan BF, Dailey LR, Mumford MD (2007) Idea evaluation: error in evaluating highly original ideas. J Creative Behav 41(1):1 (pagine esatte). https://doi.org/10.1002/j.2162-6057.2007.tb01279.x

Lillicrap TP, Santoro A, Marris L et al (2020) Backpropagation and the brain. Nat Rev Neurosci 21:335–346. https://doi.org/10.1038/s41583-020-0277-3

Lissitz RW, Willhoft JL (1985) A methodological study of the Torrance tests of creativity. J Educ Measure 22:1–11

Llano MT, d'Inverno M, Yee-King M, McCormack J, Ilsar A, Pease A, Colton S (2022) Explainable computational creativity. https://doi.org/10.48550/arXiv.2205.05682

Lubart T (2017) The 7 c's of creativity. J Creative Behav 51(4):293–296. https://doi.org/10.1002/jocb.190
May R (1959) The nature of creativity. ETC Rev General Semantics 16(3):261–276
Means D (2023) A.I. Can't write my cat story because it hasn't felt what i feel. The New York Times
Mednick SA, Mednick MT (1967) Examiner's manual: Remote Associates Test. Houghton Mifflin, Boston, MA
Miller AI (2019) The artist in the machine: the world of AI-powered creativity. MIT Press
Moruzzi C (2021) Measuring creativity: an account of natural and artificial creativity. Eur J Philos Sci 11(1):1. https://doi.org/10.1007/s13194-020-00313-w
Moruzzi C (2025) Artificial intelligence and creativity. Philos Compass 20(3):e70030. https://doi.org/10.1111/phc3.70030
Neumann E (1999) Il Sé, l'individuo, la realtà, La Biblioteca di Vivarium. Under a single title, this book brings together two insightful lectures. In: "The psyche and the levels of transformation of reality: a metapsychological attempt" and "the individual and the problem of meaning," delivered by Neumann in Ascona in 1952 and 1957 on the occasion of the Eranos conference
Newell A, Shaw JC, Simon HA (1962) The processes of creative thinking. In: Gruber HE, Terrell G, Wertheimer M (eds) Contemporary approaches to creative thinking: a symposium held at the University of Colorado. Atherton Press, pp 63–119. https://doi.org/10.1037/13117-003
Newell A, Simon HA (1961) Computer simulation of human thinking. Science 134:2011–2017. https://doi.org/10.1126/science.134.3495.2011
Newell A, Simon HA (1972) Human problem solving. Prentice-Hall
Newell A (1982) The knowledge level. Artif Intell 18(1):81–132. https://doi.org/10.1016/0004-3702(82)90012-1
Newton A, Dhole K (2023) Is AI art another industrial revolution in the making? https://doi.org/10.48550/arXiv.2301.05133
Nicholls JG (1972) Creativity in the person who will never produce anything original and useful: the concept of creativity as a normally distributed trait. Am Psychol 27(8):717–727. https://doi.org/10.1037/h0033180
Nietzsche F (2017) The will to power, penguin classics
Oh C, Song J, Choi J, Kim S, Lee S, Suh B (2018) I lead, you help but only with enough details: Understanding user experience of co-creation with artificial intelligence. In: Proceedings of the 2018 CHI conference on human factors in computing systems. ACM, pp 1–13. https://doi.org/10.1145/3173574.3174223
Ohlsson S (2012) The problems with problem solving: reflections on the rise, current status, and possible future of a cognitive research paradigm. J Probl Solving 5(1):5–19. https://doi.org/10.7771/1932-6246.1144
OpenAI (2023) GPT-4 technical report. https://doi.org/10.48550/arXiv.2303.08774
OpenAI (2023) DALL·E: creating images from text. Retrieved from https://openai.com/research/dall-e
Pedrazzi R (2021) Futuri possibili. Scenari d'arte e intelligenza artificiale [Possible futures: art and artificial intelligence scenarios]. Jaca Book, Milan
Petruzzellis F, Testolin A, Sperduti A (2024) Benchmarking GPT-4 on algorithmic problems: a systematic evaluation of prompting strategies. CoRR, abs/2402.17396
Putnam H (1960) Minds and machines. In: Hook S (ed) Dimensions of mind: a symposium. New York University Press, New York, pp 138–164
Radford A, Wu J, Child R et al (2019) Language models are unsupervised multitask learners. OpenAI Blog, 1. Retrieved from https://www.scirp.org/reference/referencespapers?referenceid=3023379
Ramesh A, Dhariwal P, Nichol A, Chu C, Chen M (2022) Hierarchical text-conditional image generation with CLIP latents. https://doi.org/10.48550/arXiv.2204.06125
Rhodes M (1961) An analysis of creativity. Phi Delta Kappan 42(7):305–310
Rosenblueth A, Wiener N (1945) The role of models in science. Philos Sci 12(4):316–321
Runco MA (2011) Runco creativity assessment battery (rCAB). Creativity Testing Services
Runco MA, Jaeger GJ (2012) The standard definition of creativity. Creat Res J 24(1):92–96. https://doi.org/10.1080/10400419.2012.650092
Runco MA (2023a) Updating the standard definition of creativity to account for the artificial creativity of AI. Creat Res J 37(1):1–5. https://doi.org/10.1080/10400419.2023.2257977
Runco MA (2023b) AI can only produce artificial creativity. J Creat 33(3):100063. https://doi.org/10.1016/j.yjoc.2023.100063
Russell S, Norvig P (2016) Artificial intelligence: a modern approach, global edition. Pearson Education
Saharia C, Chan W, Saxena S, Li L, Whang J, Denton EL, Ghasemipour SK, Gontijo Lopes R, Karagol Ayan B, Salimans T, Misra I, Norouzi M (2022) Photorealistic text-to-image diffusion models with deep language understanding. In: Proceedings of the 36th international conference on neural information processing systems (NeurIPS 2022), pp 36479–36494
Shanahan M (2023) Talking about large language models. https://doi.org/10.48550/arXiv.2212.03551
Still A, Yee-King M, d'Inverno M (2019) Creativity and feedback: Designing systems to support student learning and improve instruction. In: The future of creative learning. MIT Press, pp 1–15
Sternberg RJ, Lubart TI (1999) The concept of creativity: prospects and paradigms. In: Sternberg RJ (ed) Handbook of creativity. Cambridge University Press, pp 3–15
Sternberg RJ (2018) A triangular theory of creativity. Psychol Aesthet Creat Arts 12(1):50–67. https://doi.org/10.1037/aca0000095
Taubitz F-S, Büdenbender B, Alpers GW (2022) What the future holds: machine learning to predict success in psychotherapy. Behav Res Ther 156:104116. https://doi.org/10.1016/j.brat.2022.104116
Toivonen H, Gross O (2015) Data mining and machine learning in computational creativity. Wiley Interdiscip Rev Data Min Knowl Discov 5(6):265–275. https://doi.org/10.1002/widm.1170
Torrance EP (1966) Torrance tests of creative thinking—norms technical manual research edition—verbal tests, forms A and B—figural tests, forms A and B. Personnel Pres. Inc., Princeton
The Economist (2022) Huge foundation models are turbo-charging AI progress. The Economist. Retrivied from https://www.economist.com/interactive/briefing/2022/06/11/huge-foundation-models-are-turbo-charging-ai-progress
Varga S, Guignon C (2023) Authenticity. In: Zalta EN, Nodelman U (eds) The Stanford Encyclopedia of philosophy (Summer 2023 Edition). Retrieved from https://plato.stanford.edu/archives/sum2023/entries/authenticity/
Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser Ł, Polosukhin I (2017) Attention is all you need. Adv Neural Inf Process Syst 30:5998–6008
von Bayern AMP, Heathcote RJP, Rutz C, Kacelnik A (2009) The role of experience in problem solving and innovative tool use in crows. Curr Biol 19(22):1965–1968. https://doi.org/10.1016/j.cub.2009.10.037
von Bayern AMP, Danel S, Auersperg AMI, Mioduszewska B, Kacelnik A (2018). Compound tool construction by New Caledonian crows. Sci Rep 8:15676. https://doi.org/10.1038/s41598-018-33458-z
Wallas G (1926) The art of thought. Harcourt, Brace and Company, New York, NY

Whittington JCR, Bogacz R (2019) Theories of error back-propagation in the brain. Trends Cogn Sci 23(3):235–250. https://doi.org/10.1016/j.tics.2018.12.005

Wiggins GA (2019) A framework for description, analysis and comparison of creative systems. In: Veale T, Cardoso FA (eds) Computational creativity: the philosophy and engineering of autonomously creative systems, computational synthesis and creative systems. Springer International Publishing, Cham, 21–47

Xiao W, Chen H, Liao Q, Poggio T (2018) Biologically-plausible learning algorithms can scale to large datasets. https://doi.org/10.48550/arXiv.1811.03567

Yee-King M, d'Inverno M, Fiorucci A (2024) Designing an AI-creativity music course. In: AIMC 2024, Oxford, United Kingdom, September 9-11, 2024 (conference paper). Licensed under Creative Commons Attribution

Zhu Y, Olszewski K, Wu Y, Achlioptas P, Chai M, Yan Y, Tulyakov S (2022) Quantized GAN for complex music generation from dance videos. In: Proceedings of the European conference on computer vision (ECCV 2022). Springer, pp 182–199. https://doi.org/10.1007/978-3-031-19836-6_11

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